Rotational Source Separation with Source-Structure-Aware Loss Functions

Mathematical Formulation, Numerical Safeguards, and Implementation Guide

Technical reference for the RSS validation workflow

September 2026

Abstract

This document extends the current Rotational Source Separation (RSS) formulation from a single zero-lag correlation objective to a loss-agnostic geometric framework in which the unknown source–observer transformation is still constrained to the proper-orthogonal group SO(3), while the identification criterion is chosen according to an assumed structural property of the latent source signals. Four source-family-specific objectives are formulated for the synthetic signal families already used in the RSS validation study: (i) a spectral-overlap loss for harmonic sources, (ii) a multi-lag cross-correlation loss for filtered stochastic sources, (iii) a time–frequency overlap loss for chirp-modulated sources, and (iv) a sparse-innovation/event-overlap loss for piecewise-transient sources. The quaternion geometry, source-angle reconstruction, SO(3) parameterization, and particle-swarm optimization (PSO) search remain unchanged. The document is intended both as a mathematical reference and as a direct implementation specification for creating a second validation notebook that mirrors the current RSS experiments while replacing the separation objective with the new source-structure-aware losses.

Contents

1 Purpose, Scope, and Design Principle **4**

1.1 What changes and what does not 4

2 RSS Geometric Core to be Retained **5**

2.1 Ordered quaternion composition 5

2.2 Linear coordinate change of the common quaternion vector part 5

2.3 SO(3) parameterization 6

2.4 Candidate source-angle reconstruction 6

3 Generalized RSS Objective Interface **6**

3.1 Common numerical preprocessing 7

4 Reference Loss: Zero-Lag Correlation **7**

5 Harmonic Family: Spectral-Overlap RSS **8**
5.1 Source property . . . 8
5.2 Welch power spectral density . . . 8
5.3 Normalized spectral distributions . . . 8
5.4 Pairwise spectral overlap . . . 8
5.5 Diagnostics to store . . . 9
5.6 Failure modes . . . 9

6 Filtered-Noise Family: Multi-Lag Cross-Correlation RSS **9**
6.1 Source property . . . 9
6.2 Normalized cross-correlation at lag ℓ . . . 9
6.3 Lag weights . . . 10
6.4 Multi-lag loss . . . 10
6.5 Diagnostics . . . 10

7 Chirp-Modulated Family: Time–Frequency Overlap RSS **11**
7.1 Source property . . . 11
7.2 Short-time Fourier transform . . . 11
7.3 Per-frame normalized spectral distributions . . . 11
7.4 Time–frequency overlap . . . 11
7.5 Optional energy weighting . . . 12
7.6 Ridge separation as a diagnostic, not the primary loss . . . 12

8 Piecewise-Transient Family: Sparse-Innovation RSS **12**
8.1 Source property . . . 12
8.2 Second-difference innovations . . . 13
8.3 Hoyer sparsity . . . 13
8.4 Pairwise event-overlap loss . . . 13
8.5 Combined piecewise loss . . . 13
8.6 Interpretation and limitations . . . 13

9 Unified Family-Specific Objective **14**

10 PSO Integration **14**
10.1 Generalized objective-evaluation pseudocode . . . 15

11 Loss-Landscape Validation Before Full PSO Experiments **15**

12 Experimental Protocol for the Second Validation Notebook **16**

12.1 Paired comparison principle 16

12.2 Required run modes 16

12.3 Preserve all existing evaluation metrics 16

12.4 Strongly recommended additional diagnostics 16

13 Suggested Notebook Organization for Codex **17**

14 Recommended Software Interface **18**

15 Validation Criteria for Deciding Whether a Specialized Loss is Useful **19**

16 Relationship Between the Four Losses **20**

17 Compact Algorithm Summary **20**

18 Implementation Checklist for Codex **20**

19 Notes on Interpretation for the Journal Paper **21**

A Family Definitions from the Current RSS Synthetic Study **21**

A.1 Harmonic 21

A.2 Filtered noise 21

A.3 Chirp modulated 22

A.4 Piecewise transient 22

B Notation Summary **22**

C Source Basis and Explicit Unknowns **23**

1 Purpose, Scope, and Design Principle

The current RSS formulation separates latent rotational components by searching for the unknown coordinate transformation between an observer frame and a source frame. The key geometric result is independent of the statistical property later used to identify the correct transformation: after the compounded physical rotation has been represented by a quaternion, the *vector part* of that common quaternion changes coordinates linearly between the two right-handed orthonormal frames.

The current implementation uses mutual zero-lag uncorrelatedness as its identification property. The corresponding loss is effective when the unknown source-angle signals are sufficiently uncorrelated, but zero-lag correlation is not intended to be a universal property of every possible rotational source. The more general RSS viewpoint adopted here is therefore

$$\boxed{\text{RSS geometry} + \text{source-appropriate identification loss}} \tag{1}$$

rather than defining RSS itself as a correlation-only method.

For a trial source–observer transformation $A \in \mathrm{SO}(3)$, let $\mathbf{S}(A)$ denote the complete set of reconstructed source-angle signals. RSS can then be written generically as

$$\boxed{A^\star = \arg \min_{A \in \mathrm{SO}(3)} \mathcal{L}_{\mathcal{P}}(\mathbf{S}(A)) \,,} \tag{2}$$

where $\mathcal{P}$ is an assumed source property and $\mathcal{L}_{\mathcal{P}}$ is a loss that is small when the reconstructed signals satisfy that property.

1.1 What changes and what does not

The following components **must remain identical to the current RSS implementation**:

- observer-angle to quaternion conversion;
- the fixed rotation order used in quaternion composition;
- the SO(3) source–observer coordinate transform;
- Rodrigues/exponential parameterization of the separation matrix;
- quaternion-to-source-angle reconstruction and its branch convention;
- the PSO implementation, initialization policy, stopping budget, and boundary handling used by the existing code;
- the synthetic data generation, random frame generation, evaluation metrics, and experiment bookkeeping in the current validation notebook.

Only the scalar objective evaluated after reconstructing the candidate source-angle matrix is to be replaced.

Important implementation constraint. The current validation notebook itself was not supplied with the source documents used to prepare this reference. Therefore, notebook-specific experiments that are not described in the RSS manuscript cannot be enumerated here. When this document is given to Codex, the new notebook should mirror the existing validation notebook cell-for-cell and experiment-for-experiment wherever possible, while replacing the objective with the loss dispatch described in this document. No existing experiment should be silently removed or redesigned.

2 RSS Geometric Core to be Retained

2.1 Ordered quaternion composition

For a rotation of angle $\theta_{u_i}(t)$ about unit axis $\hat{\boldsymbol{u}}_i$, define

$$q_{u_i}(t) = \cos\left(\frac{\theta_{u_i}(t)}{2}\right) + \hat{\boldsymbol{u}}_i \sin\left(\frac{\theta_{u_i}(t)}{2}\right). \tag{3}$$

The RSS formulation uses the fixed composition order

$$q_Q(t) = q_{u_3}(t) \otimes q_{u_2}(t) \otimes q_{u_1}(t). \tag{4}$$

Writing

$$q_Q(t) = w_u(t) + x_u(t)\hat{\boldsymbol{u}}_1 + y_u(t)\hat{\boldsymbol{u}}_2 + z_u(t)\hat{\boldsymbol{u}}_3, \tag{5}$$

and defining

$$C_i = \cos\left(\frac{\theta_{u_i}}{2}\right), \qquad S_i = \sin\left(\frac{\theta_{u_i}}{2}\right), \tag{6}$$

produces

$$w_u = C_3C_2C_1 + S_3S_2S_1, \tag{7}$$
$$x_u = C_3C_2S_1 - S_3S_2C_1, \tag{8}$$
$$y_u = C_3S_2C_1 + S_3C_2S_1, \tag{9}$$
$$z_u = S_3C_2C_1 - C_3S_2S_1. \tag{10}$$

Exactly the same convention must be used for the observer angles and for candidate source-angle reconstruction.

2.2 Linear coordinate change of the common quaternion vector part

The source and observer descriptions represent the same physical resultant quaternion. Let

$$\boldsymbol{v}_R(t) = \begin{bmatrix} x_R(t) & y_R(t) & z_R(t) \end{bmatrix}^T, \qquad \boldsymbol{v}_S(t) = \begin{bmatrix} x_S(t) & y_S(t) & z_S(t) \end{bmatrix}^T. \tag{11}$$

Then

$$\boxed{vect v_S(t) = A\boldsymbol{v}_R(t), \qquad A \in \mathrm{SO}(3),} \tag{12}$$

and the scalar quaternion component is frame invariant,

$$w_S(t) = w_R(t). \tag{13}$$

If $\mathbf{S}_b = [\hat{\boldsymbol{s}}_1\ \hat{\boldsymbol{s}}_2\ \hat{\boldsymbol{s}}_3]$ and $\mathbf{R}_b = [\hat{\boldsymbol{r}}_1\ \hat{\boldsymbol{r}}_2\ \hat{\boldsymbol{r}}_3]$ are the right-handed orthonormal source and observer bases, then

$$A = \mathbf{S}_b^T \mathbf{R}_b, \tag{14}$$

which directly yields $A^TA = I$ and $\det A = 1$.

2.3 SO(3) parameterization

The current RSS optimizer represents the unknown transformation by a rotation vector

$$\boldsymbol{\omega} = \begin{bmatrix} \omega_x & \omega_y & \omega_z \end{bmatrix}^T, \tag{15}$$

with skew-symmetric cross-product matrix

$$[\boldsymbol{\omega}]_\times = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}. \tag{16}$$

The separation matrix is

$$A(\boldsymbol{\omega}) = \exp([\boldsymbol{\omega}]_\times). \tag{17}$$

With $\theta = \|\boldsymbol{\omega}\|_2$, Rodrigues' formula gives

$$A(\boldsymbol{\omega}) = I_3 + \frac{\sin\theta}{\theta}[\boldsymbol{\omega}]_\times + \frac{1-\cos\theta}{\theta^2}[\boldsymbol{\omega}]_\times^2. \tag{18}$$

The implementation should use the numerical limits

$$\frac{\sin\theta}{\theta} \to 1, \qquad \frac{1-\cos\theta}{\theta^2} \to \frac{1}{2}, \qquad \theta \to 0, \tag{19}$$

to avoid division by zero.

2.4 Candidate source-angle reconstruction

For each candidate $\boldsymbol{\omega}$ and each time sample,

$$\boldsymbol{v}_S(t;\boldsymbol{\omega}) = A(\boldsymbol{\omega})\boldsymbol{v}_R(t), \qquad w_S(t;\boldsymbol{\omega}) = w_R(t). \tag{20}$$

The source angles corresponding to the adopted $1 \to 2 \to 3$ convention are reconstructed using

$$\theta_{S_1}(t;\boldsymbol{\omega}) = \operatorname{atan2}\big(2(w_S x_S + y_S z_S), 1 - 2(x_S^2 + y_S^2)\big), \tag{21}$$

$$\theta_{S_2}(t;\boldsymbol{\omega}) = \sin^{-1}\Big(\operatorname{clip}_{[-1,1]}\{2(w_S y_S - z_S x_S)\}\Big), \tag{22}$$

$$\theta_{S_3}(t;\boldsymbol{\omega}) = \operatorname{atan2}\big(2(w_S z_S + x_S y_S), 1 - 2(y_S^2 + z_S^2)\big). \tag{23}$$

Stack the candidate signals as

$$\mathbf{S}(\boldsymbol{\omega}) = \begin{bmatrix} \theta_{S_1}(1) & \cdots & \theta_{S_1}(N) \\ \theta_{S_2}(1) & \cdots & \theta_{S_2}(N) \\ \theta_{S_3}(1) & \cdots & \theta_{S_3}(N) \end{bmatrix} \in \mathbb{R}^{3\times N}. \tag{24}$$

Every loss in the following sections receives this same matrix as its primary input.

3 Generalized RSS Objective Interface

The generic PSO fitness is

$$\boxed{J(\boldsymbol{\omega};\mathcal{P}) = \mathcal{L}_{\mathcal{P}}\left(\mathbf{S}(\boldsymbol{\omega}); f_s, \boldsymbol{\eta}_{\mathcal{P}}\right),} \tag{25}$$

where f_s is the sampling frequency and $\boldsymbol{\eta}_{\mathcal{P}}$ contains loss-specific hyperparameters. The optimized transform is

$$\boldsymbol{\omega}^\star = \arg\min_{\boldsymbol{\omega}\in\Omega} J(\boldsymbol{\omega};\mathcal{P}), \qquad \Omega = [-\pi,\pi]^3 \tag{26}$$

for the current RSS search domain, followed by

$$A^\star = A(\boldsymbol{\omega}^\star), \qquad \widehat{\mathbf{S}} = \mathbf{S}(\boldsymbol{\omega}^\star). \tag{27}$$

3.1 Common numerical preprocessing

Before computing a source-structure loss, apply the following operations consistently.

Finite-value guard. If any candidate angle is NaN or infinite, return a deliberately poor finite objective value, e.g. $J_{\text{bad}} = 10^6$.

Temporal continuity. For spectral, time–frequency, and derivative-based losses, discontinuous 2π angle wrapping can create artificial high-frequency content. Use the same continuity handling already adopted by the current implementation. If no continuity step exists and the candidate angles can cross the principal branch, apply row-wise phase unwrapping only for *loss evaluation*; retain the original reconstructed angle convention for reported physical outputs. This choice must be recorded because it changes the objective.

Mean removal. For losses based on spectral shape, cross-correlation, or innovations, center each candidate row:

$$\widetilde{s}_i[n] = s_i[n] - \frac{1}{N}\sum_{n=0}^{N-1} s_i[n]. \tag{28}$$

Degenerate variance guard. Let

$$\sigma_i^2 = \frac{1}{N}\sum_n \widetilde{s}_i[n]^2. \tag{29}$$

If $\sigma_i^2 < \varepsilon_\sigma$ for any source, return J_{bad}. Use a numerical floor such as $\varepsilon_\sigma = 10^{-12}$ in squared-radian units unless the existing implementation already defines a different tolerance. This guard prevents undefined normalization; it is not intended to enforce equal source amplitudes.

4 Reference Loss: Zero-Lag Correlation

The existing RSS objective should remain available as a baseline and as a regression test for the refactored implementation. Let the centered candidate matrix be $\overline{\mathbf{S}}$. Define

$$\mathbf{C}_S = \frac{1}{N}\overline{\mathbf{S}}\,\overline{\mathbf{S}}^T, \qquad \mathbf{D} = \mathrm{Diag}(C_{11}, C_{22}, C_{33}), \tag{30}$$

and

$$\mathbf{R}_S = \mathbf{D}^{-1/2}\mathbf{C}_S\mathbf{D}^{-1/2}. \tag{31}$$

The current loss is

$$\boxed{\mathcal{L}_{\text{corr}} = \frac{1}{3}\left\|\mathrm{offdiag}(\mathbf{R}_S)\right\|_F^2 = \frac{2}{3}\left(\rho_{12}^2 + \rho_{13}^2 + \rho_{23}^2\right).} \tag{32}$$

This loss requires only a second-order zero-lag source assumption and no knowledge of the source family.

5 Harmonic Family: Spectral-Overlap RSS

5.1 Source property

For harmonic sources whose dominant spectral content is sufficiently distinct, a correct separation should concentrate each recovered source into its own dominant frequency lobes rather than distributing the same lobes across multiple outputs. This motivates minimizing pairwise overlap between normalized power spectra.

This is a **source-structure assumption**, not a universal property of harmonic signals. If two true sources contain identical or strongly overlapping frequencies, spectral overlap alone cannot uniquely identify the physical source frame. The proposed loss must therefore be validated on the actual generated cases before being treated as an identifying criterion.

5.2 Welch power spectral density

For centered candidate source $\widetilde{s}_i[n]$, estimate the one-sided power spectral density (PSD) using Welch averaging:

$$P_i[f_k] = \mathrm{WelchPSD}\{\widetilde{s}_i[n]\}. \tag{33}$$

Restrict the PSD to the analysis band $\mathcal{F}$ relevant to the simulated harmonic content. For the current synthetic RSS signals, an initial choice of

$$\mathcal{F} = [0, 5] \text{ Hz} \tag{34}$$

is sufficient to contain the documented harmonic and chirp frequency ranges at $f_s = 100$ Hz. Keep this band configurable rather than hard-coded into the core loss function.

A practical initial Welch configuration is

$$n_{\mathrm{seg}} = \min(256, N), \qquad n_{\mathrm{overlap}} = \lfloor 0.5 n_{\mathrm{seg}} \rfloor, \tag{35}$$

using a Hann window and an FFT length at least n_{seg} (e.g. the next power of two). The same frequency grid must be used for all three candidate outputs.

5.3 Normalized spectral distributions

Convert each PSD into a unit-mass spectral distribution:

$$\boxed{p_i[k] = \frac{(P_i[f_k] + \varepsilon_P)^{\gamma_H}}{\sum_{k \in \mathcal{F}} (P_i[f_k] + \varepsilon_P)^{\gamma_H}}} \tag{36}$$

with small $\varepsilon_P > 0$. The exponent $\gamma_H \geq 1$ controls how strongly dominant lobes are emphasized. Recommended starting values are $\gamma_H = 1$ for a pure power-distribution overlap and $\gamma_H = 2$ as a sensitivity test that emphasizes dominant peaks.

5.4 Pairwise spectral overlap

Use the Bhattacharyya coefficient between spectral distributions:

$$B_{ij}^{(H)} = \sum_{k \in \mathcal{F}} \sqrt{p_i[k] p_j[k]}. \tag{37}$$

For normalized nonnegative distributions,

$$0 \le B_{ij}^{(H)} \le 1. \tag{38}$$

Values near 0 indicate distinct spectral support; values near 1 indicate very similar spectral distributions.

The proposed harmonic RSS loss is therefore

$$\boxed{\mathcal{L}_{\mathrm{H}} = \frac{1}{3}\left(B_{12}^{(H)} + B_{13}^{(H)} + B_{23}^{(H)}\right).} \tag{39}$$

The objective is minimized.

5.5 Diagnostics to store

For every final solution, save

$$B_{12}^{(H)}, \quad B_{13}^{(H)}, \quad B_{23}^{(H)}, \quad \mathcal{L}_{\mathrm{H}}, \tag{40}$$

and the PSDs of the final three recovered sources. For representative examples, plot source-vs-reconstruction PSDs in addition to the existing time-domain overlays.

5.6 Failure modes

Spectral-overlap RSS can fail or become non-identifying when:

- two true sources share the same dominant frequencies;
- harmonics from one source fall on fundamentals/harmonics of another source;
- the observation interval is too short for adequate spectral resolution;
- branch discontinuities introduce artificial broadband energy;
- a candidate solution artificially narrows spectra without corresponding to the true frame.

The loss-landscape validation in Section 11 is therefore mandatory before interpreting improvements as physical identification.

6 Filtered-Noise Family: Multi-Lag Cross-Correlation RSS

6.1 Source property

The current filtered-noise family is stochastic and temporally correlated within each individual source. For mutually generated stochastic sources, correct separation is expected to reduce *cross*-dependence between different recovered sources. Because the individual filtered-noise signals have memory, a multi-lag criterion uses more second-order information than zero-lag correlation alone while remaining a correlation-based blind criterion.

6.2 Normalized cross-correlation at lag ℓ

For centered signals $\widetilde{s}_i[n]$ and $\widetilde{s}_j[n]$, define for integer lag $\ell \ge 0$

$$\rho_{ij}[\ell] = \frac{\sum_{n=0}^{N-1-\ell} \widetilde{s}_i[n]\widetilde{s}_j[n+\ell]}{\sqrt{\sum_{n=0}^{N-1-\ell} \widetilde{s}_i[n]^2}\sqrt{\sum_{n=0}^{N-1-\ell} \widetilde{s}_j[n+\ell]^2 + \varepsilon_\rho}}. \tag{41}$$

For negative lags use the corresponding swapped alignment. The implementation may equivalently use a standard normalized cross-correlation routine provided that normalization is consistent across all lags and cases.

6.3 Lag weights

Choose a maximum lag L and nonnegative symmetric weights $w_\ell = w_{-\ell}$. A simple triangular choice is

$$w_\ell = 1 - \frac{|\ell|}{L+1}, \qquad -L \le \ell \le L. \tag{42}$$

Normalize

$$\bar{w}_\ell = \frac{w_\ell}{\sum_{r=-L}^{L} w_r}. \tag{43}$$

For the current $f_s = 100$ Hz simulations, a useful starting point is

$$L = \min\left(100, \left\lfloor \frac{N}{4} \right\rfloor\right), \tag{44}$$

which evaluates up to approximately one second of lag for the documented sample rate. This is an implementation starting point rather than a theoretically optimal universal value.

6.4 Multi-lag loss

Define the pairwise lag-energy

$$C_{ij}^{(F)} = \sum_{\ell=-L}^{L} \bar{w}_\ell \rho_{ij}^2[\ell]. \tag{45}$$

The filtered-noise specialized loss is

$$\boxed{\mathcal{L}_{\mathrm{F}} = \frac{1}{3}\left(C_{12}^{(F)} + C_{13}^{(F)} + C_{23}^{(F)}\right).} \tag{46}$$

Because each squared correlation is in $[0, 1]$ and the lag weights sum to one, $\mathcal{L}_{\mathrm{F}}$ is also bounded between 0 and 1.

The existing zero-lag correlation loss is recovered as the special case $L = 0$ (up to the constant convention used in Eq. (32)). This provides a direct regression test.

6.5 Diagnostics

Save the pairwise cross-correlation functions at the final solution and the scalar values

$$C_{12}^{(F)}, \quad C_{13}^{(F)}, \quad C_{23}^{(F)}, \quad \mathcal{L}_{\mathrm{F}}. \tag{47}$$

A useful diagnostic plot is the mean absolute cross-correlation magnitude versus lag before and after RSS.

7 Chirp-Modulated Family: Time–Frequency Overlap RSS

7.1 Source property

For frequency-varying sources with sufficiently distinct instantaneous-frequency trajectories, correct separation should reduce the sharing of the same time–frequency energy ridges across multiple recovered components. A full time–frequency overlap loss is preferred over a pure ridge-distance objective because ridge extraction can be unstable for partially mixed candidate signals and because simply maximizing ridge distance may reward physically incorrect transformations that exaggerate separation.

7.2 Short-time Fourier transform

For each centered candidate source, compute

$$X_i[k,m] = \sum_n \widetilde{s}_i[n] h[n - mH] e^{-j2\pi kn/N_{\mathrm{FFT}}}, \tag{48}$$

where $h[\cdot]$ is a Hann window, H is the hop size, k is the frequency-bin index, and m is the time-frame index. Let

$$P_i[k,m] = |X_i[k,m]|^2. \tag{49}$$

Restrict to the frequency band $\mathcal{F}_{TF}$ containing the simulated chirp content. A starting choice for the current validation data is again $[0, 5]$ Hz.

For $f_s = 100$ Hz and $N \in \{720, 840, 960, 1080\}$, a practical starting STFT configuration is

$$n_{\mathrm{seg}} = 128, \qquad n_{\mathrm{overlap}} = 96, \qquad H = 32, \qquad N_{\mathrm{FFT}} = 256. \tag{50}$$

These values should be kept configurable; the objective should not depend on hidden notebook constants.

7.3 Per-frame normalized spectral distributions

For each time frame m, normalize the frequency energy:

$$\boxed{p_i[k,m] = \frac{P_i[k,m] + \varepsilon_{TF}}{\sum_{k \in \mathcal{F}_{TF}} (P_i[k,m] + \varepsilon_{TF})}.} \tag{51}$$

Thus $\sum_k p_i[k,m] = 1$ for every valid frame.

7.4 Time–frequency overlap

At each frame, define

$$B_{ij}^{(C)}[m] = \sum_{k \in \mathcal{F}_{TF}} \sqrt{p_i[k,m] p_j[k,m]}. \tag{52}$$

Average across time:

$$\overline{B}_{ij}^{(C)} = \frac{1}{M} \sum_{m=1}^{M} B_{ij}^{(C)}[m]. \tag{53}$$

The chirp specialized loss is

$$\boxed{\mathcal{L}_{\mathrm{C}} = \frac{1}{3}\left(\overline{B}_{12}^{(C)} + \overline{B}_{13}^{(C)} + \overline{B}_{23}^{(C)}\right).} \tag{54}$$

This objective is bounded in $[0, 1]$ and is minimized when the candidate sources occupy distinct frequency regions at each time frame.

7.5 Optional energy weighting

If low-energy edge frames produce unstable normalized spectra, define a common nonnegative frame weight from the total candidate energy

$$E[m] = \sum_{i=1}^{3} \sum_{k \in \mathcal{F}_{TF}} P_i[k, m], \qquad \alpha_m = \frac{E[m]}{\sum_r E[r] + \varepsilon_E}, \tag{55}$$

and replace the time average by

$$\overline{B}_{ij}^{(C)} = \sum_m \alpha_m B_{ij}^{(C)}[m]. \tag{56}$$

The initial implementation should first use the unweighted version for transparency; enable energy weighting only if edge-frame instability is observed.

7.6 Ridge separation as a diagnostic, not the primary loss

A soft instantaneous-frequency centroid can be calculated as

$$\widehat{f}_i[m] = \sum_{k \in \mathcal{F}_{TF}} f_k p_i[k, m]. \tag{57}$$

A diagnostic mean pairwise ridge distance is

$$D_{\mathrm{ridge}} = \frac{1}{3M f_{\mathrm{Nyq}}} \sum_{i<j} \sum_m \left| \widehat{f}_i[m] - \widehat{f}_j[m] \right|. \tag{58}$$

A larger D_{ridge} is consistent with more separated time-varying frequency content, but it should not be used as the primary optimization target until it is shown that maximizing it identifies the true physical frame rather than simply exaggerating ridge distance.

8 Piecewise-Transient Family: Sparse-Innovation RSS

8.1 Source property

The current piecewise-transient family combines a piecewise-interpolated background with localized Gaussian bursts. Orthogonal mixing tends to distribute a change or burst originating in one latent source across several observed components. A plausible source prior is therefore that, after correct separation, each recovered signal has relatively sparse temporal innovations and that innovation events are less simultaneously shared between different sources.

The appropriate innovation operator depends on the generator. Because the documented background is linearly interpolated, the *second difference* is the natural default: it suppresses locally linear trends while responding strongly near slope changes and localized transient curvature.

8.2 Second-difference innovations

For each centered source, define

$$d_i[n] = \Delta^2 \widetilde{s}_i[n] = \widetilde{s}_i[n+1] - 2\widetilde{s}_i[n] + \widetilde{s}_i[n-1], \qquad n = 1, \ldots, N-2. \tag{59}$$

Define the nonnegative event envelope

$$e_i[n] = |d_i[n]|. \tag{60}$$

8.3 Hoyer sparsity

For a nonzero innovation vector $\boldsymbol{d}_i \in \mathbb{R}^{N_d}$, define Hoyer sparsity

$$H_i = \frac{\sqrt{N_d} - \|\boldsymbol{d}_i\|_1 / (\|\boldsymbol{d}_i\|_2 + \varepsilon_d)}{\sqrt{N_d} - 1}. \tag{61}$$

For nonnegative sparsity interpretation, $H_i \in [0, 1]$, with larger values representing more concentrated/sparse innovations. Define

$$\mathcal{L}_{\text{sparse}} = 1 - \frac{1}{3} \sum_{i=1}^{3} H_i. \tag{62}$$

Thus lower loss corresponds to sparser second-difference activity.

8.4 Pairwise event-overlap loss

Normalize each event envelope using its Euclidean norm and compute

$$O_{ij}^{(P)} = \frac{\sum_n e_i[n] e_j[n]}{\|\boldsymbol{e}_i\|_2 \|\boldsymbol{e}_j\|_2 + \varepsilon_e}. \tag{63}$$

This is a cosine similarity between nonnegative event envelopes and lies in $[0, 1]$. Define

$$\mathcal{L}_{\text{event}} = \frac{1}{3} \left(O_{12}^{(P)} + O_{13}^{(P)} + O_{23}^{(P)} \right). \tag{64}$$

8.5 Combined piecewise loss

The recommended initial loss combines individual innovation sparsity with pairwise event separation:

$$\boxed{\mathcal{L}_{\text{P}} = \lambda_P \mathcal{L}_{\text{sparse}} + (1 - \lambda_P) \mathcal{L}_{\text{event}}, \qquad 0 \le \lambda_P \le 1.} \tag{65}$$

Use $\lambda_P = 0.5$ as the first implementation setting and test $\lambda_P \in \{0.25, 0.5, 0.75\}$ as a small sensitivity analysis.

8.6 Interpretation and limitations

The sparse-innovation criterion is appropriate only when the latent sources actually have localized or sparse changes. If true sources frequently change at the same times, event overlap is not expected to vanish. If the piecewise generator produces broad smooth components rather than localized innovation activity, the sparsity term may be more informative than the event-overlap term. Both terms should therefore be examined separately in diagnostics before relying on the combined objective.

9 Unified Family-Specific Objective

For the four existing synthetic families, the proposed objective dispatch is

$$\mathcal{L}_{\text{family}}(\mathbf{S}) = \begin{cases} \mathcal{L}_{\text{H}}, & \text{harmonic,} \\ \mathcal{L}_{\text{F}}, & \text{filtered noise,} \\ \mathcal{L}_{\text{C}}, & \text{chirp modulated,} \\ \mathcal{L}_{\text{P}}, & \text{piecewise transient.} \end{cases} \tag{66}$$

Table 1: Recommended initial specialized-loss settings for the current RSS synthetic validation. All settings should be exposed through a configuration object rather than hard-coded.

Family	Primary loss	Initial settings
Harmonic	PSD Bhattacharyya overlap	Welch Hann; $n_{\text{seg}} = \min(256, N)$; 50% overlap; 0–5 Hz; $\gamma_H = 1$
Filtered noise	Multi-lag squared cross-correlation	$L = \min(100, \lfloor N/4 \rfloor)$; triangular symmetric weights
Chirp modulated	STFT time–frequency Bhattacharyya overlap	Hann; $n_{\text{seg}} = 128$; overlap 96; $N_{\text{FFT}} = 256$; 0–5 Hz
Piecewise transient	Hoyer sparsity + event-envelope overlap	second difference; $\lambda_P = 0.5$

10 PSO Integration

The PSO state and search procedure should remain the same as the current RSS implementation. For particle p at iteration k, retain

$$\boldsymbol{v}_p^{(k+1)} = w\boldsymbol{v}_p^{(k)} + c_1\boldsymbol{r}_{1,p}^{(k)} \odot \left(\boldsymbol{b}_p^{(k)} - \boldsymbol{\omega}_p^{(k)}\right) + c_2\boldsymbol{r}_{2,p}^{(k)} \odot \left(\boldsymbol{g}^{(k)} - \boldsymbol{\omega}_p^{(k)}\right), \tag{67}$$

with the same current settings reported in the RSS formulation:

$$P = 80, \qquad K_{\max} = 1000, \qquad w = 0.9, \qquad c_1 = c_2 = 1.5. \tag{68}$$

The current formulation also describes an exponentially reduced position step from 10^{-1} to 10^{-3} and one fixed proper-orthogonal initialization together with uniformly sampled remaining particles.

Do not invent missing PSO details. The supplied mathematical formulation explicitly notes that the exact out-of-bounds rule for the $[-\pi, \pi]^3$ search domain was not specified in the manuscript. The new implementation must therefore inspect and preserve the boundary rule in the existing RSS code (clipping, reflection, wrapping, or other behavior) rather than replacing it with a new choice. The same applies to the exact representation of the fixed seed A_0 and the exact exponentially decaying step formula if these are already implemented in the current notebook/codebase.

10.1 Generalized objective-evaluation pseudocode

```
def rss_objective(omega, observer_quaternion_data, family, fs, loss_cfg):
    A = so3_from_rotation_vector(omega) # unchanged RSS geometry
    S = reconstruct_source_angles(A, observer_quaternion_data)

    if not finite(S):
        return BAD_LOSS

    S_loss = continuity_handling_for_loss(S, loss_cfg)
    S_loss = row_center(S_loss)

    if any(row_variance(S_loss) < variance_floor):
        return BAD_LOSS

    if family == "harmonic":
        return harmonic_spectral_overlap_loss(S_loss, fs, loss_cfg)
    elif family == "filtered_noise":
        return multilag_crosscorr_loss(S_loss, fs, loss_cfg)
    elif family == "chirp_modulated":
        return time_frequency_overlap_loss(S_loss, fs, loss_cfg)
    elif family == "piecewise_transient":
        return sparse_innovation_loss(S_loss, fs, loss_cfg)
    else:
        raise ValueError("Unsupported specialized-loss family")
```

The baseline correlation objective should be retained as an additional selectable option such as `loss_mode="corr"` so that the refactor can reproduce the original RSS results before specialized runs are attempted.

11 Loss-Landscape Validation Before Full PSO Experiments

Before running all 200 cases with a specialized loss, verify that the proposed loss actually prefers the true physical transformation for representative cases. This is particularly important for spectral and time–frequency losses because a signal property that sounds plausible qualitatively is not automatically an identifying property of the current synthetic generator.

For each selected case:

1. store the known true source–observer transformation A_{true};
2. evaluate $\mathcal{L}_{\text{family}}(\mathbf{S}(A_{\text{true}}))$;
3. sample M_R random candidate rotations $A_r \in \mathrm{SO}(3)$ using the same random-rotation routine used elsewhere in the project;
4. reconstruct $\mathbf{S}(A_r)$ and evaluate the specialized loss for every candidate;
5. calculate the geodesic frame error

$$d_{\mathrm{SO}(3)}(A_r, A_{\text{true}}) = \cos^{-1}\left(\mathrm{clip}_{[-1,1]}\,\frac{\mathrm{tr}(A_r^T A_{\text{true}}) - 1}{2}\right); \tag{69}$$

6. plot loss versus $d_{\mathrm{SO}(3)}$ and compute the percentile/rank of the true-frame loss among the random candidates.

A promising identifying loss should usually assign low loss to transformations near the true frame. It is not necessary for the landscape to be globally convex; PSO is already used because the RSS objective is nonconvex. However, if low-loss candidates are systematically far from A_{true}, the

loss is enforcing the intended signal property without identifying the physical source frame and should not be used as the main RSS objective.

Recommended initial landscape test size:

$$M_R = 2000\text{–}10000 \tag{70}$$

random rotations for a small number of representative cases from each family. This test is inexpensive compared with repeated 1000-iteration PSO runs and should be completed first.

12 Experimental Protocol for the Second Validation Notebook

12.1 Paired comparison principle

The specialized-loss experiments should be performed on **exactly the same synthetic cases and source–observer geometries** used by the current RSS validation. Do not regenerate a different dataset for the specialized method if the existing cases can be reused. A paired design removes case-to-case variability when comparing correlation-based RSS with specialized-loss RSS.

The documented synthetic study contains 200 cases, 50 per family, sampled at $f_s = 100$ Hz, with $N \in \{720, 840, 960, 1080\}$ and three latent rotations per case. The new notebook should preserve these settings and all current family generators, amplitude tiers, whitening/rescaling rules, and random frame transformations.

12.2 Required run modes

At minimum, implement the following selectable run modes:

1. **Regression mode:** original zero-lag correlation objective. Results should reproduce the current notebook within numerical tolerance.
2. **Specialized mode:** family-specific objective from Eq. (66).
3. **Optional cross-loss mode:** run a nonmatching loss on another family for diagnostic purposes. This is useful for determining whether a specialized loss is genuinely source-structure-specific or simply a stronger universal objective.

12.3 Preserve all existing evaluation metrics

Whatever metrics and plots are already used by the current validation notebook should be retained. Based on the supplied RSS manuscript, the minimum existing validation metrics are:

- absolute source correlation with ground truth;
- angular RMSE in radians;
- family-level and overall performance distributions;
- representative source-reconstruction plots.

If the current notebook contains additional metrics, figures, case filters, or export files, the specialized-loss notebook should reproduce them as well.

12.4 Strongly recommended additional diagnostics

The following quantities are especially useful for understanding the new objectives and require little additional computation:

- final specialized objective value;
- the pairwise components of that objective;
- SO(3) geodesic frame error from Eq. (69) for synthetic data;
- PSO convergence trace (best loss versus iteration);
- runtime per case;
- whether the run reached a configurable reconstruction-success threshold;
- loss-specific diagnostic plots: PSD overlap, lagged cross-correlation, spectrogram overlap, or innovation envelopes.

These diagnostics need not all appear in the four-page journal Letter. Their purpose is to determine whether the specialized losses are scientifically useful enough to justify inclusion.

13 Suggested Notebook Organization for Codex

Create a second notebook rather than modifying the original validation notebook. A suitable name is

`RSS_Validation_Specialized_Losses.ipynb`.

The notebook should use the existing modular RSS code wherever possible. The intended structure is:

Notebook Section 1: Imports and configuration

- import the same RSS simulation, quaternion, PSO, plotting, and evaluation modules used by the original notebook;
- add a dedicated loss-configuration dataclass/dictionary;
- define one master random seed policy and keep source-generation seeds separate from PSO seeds if the current code already does so.

Notebook Section 2: Regression check

Run a small subset with `loss_mode="corr"` and verify that outputs agree with the original notebook. Do not proceed to interpret specialized-loss results until this check passes.

Notebook Section 3: Loss-landscape diagnostics

For representative cases from each family, evaluate the true frame and many random rotations. Produce:

- loss-vs-frame-error scatter plots;
- histogram of random-candidate loss values with the true-frame loss marked;
- percentile rank of the true-frame loss;
- family-specific diagnostic visualization.

Notebook Section 4: Full specialized-loss validation

Run all cases currently used in the RSS simulation study using the mapped family-specific loss. Save results using the same case identifiers as the original notebook.

Notebook Section 5: Paired baseline-versus-specialized comparison

For each family, compare the existing correlation-RSS output with the specialized-loss output on the same cases. Recommended summaries are paired differences in:

$$\Delta|\rho| = |\rho|_{\text{specialized}} - |\rho|_{\text{corr}}, \tag{71}$$

$$\Delta\text{RMSE} = \text{RMSE}_{\text{specialized}} - \text{RMSE}_{\text{corr}}, \tag{72}$$

and

$$\Delta d_{\text{SO(3)}} = d_{\text{SO(3),specialized}} - d_{\text{SO(3),corr}}. \tag{73}$$

Negative ΔRMSE and $\Delta d_{\text{SO(3)}}$ indicate improvement; positive $\Delta|\rho|$ indicates improvement.

Notebook Section 6: Representative visualizations

For each family, show at least one successful, median, and difficult case where practical. Include the original time-domain comparison plus the relevant structural diagnostic:

- harmonic: normalized PSDs;
- filtered noise: pairwise cross-correlation vs lag;
- chirp: STFT/spectrograms and optional ridge-centroid trajectories;
- piecewise transient: second-difference/event envelopes.

Notebook Section 7: Hyperparameter sensitivity

Keep this compact. Suggested first sweeps are:

- harmonic: $\gamma_H \in \{1, 2\}$;
- filtered noise: $L \in \{25, 50, 100\}$ samples;
- chirp: $n_{\text{seg}} \in \{96, 128, 192\}$ with approximately 75% overlap;
- piecewise: $\lambda_P \in \{0.25, 0.5, 0.75\}$.

Run these on a representative subset first. Do not multiply the full PSO study by every hyperparameter combination unless the subset shows meaningful sensitivity.

Notebook Section 8: Export

Export both case-level and aggregate results to structured CSV/NPZ/JSON files using the same identifiers as the original study. Include loss configuration metadata in every saved result so that a plot cannot become detached from the objective that produced it.

14 Recommended Software Interface

A modular implementation should separate RSS geometry from source-identification criteria.

```
# Geometry layer: must not know the loss family.
def reconstruct_candidate_sources(omega, observer_state, rotation_cfg):
    ...
    return S_candidate

# Loss layer: receives already reconstructed source angles.
```

```
def compute_rss_loss(S_candidate, fs, loss_mode, loss_cfg):
    ...
    return scalar_loss, diagnostics

# Optimizer layer: calls the two layers.
def fit_rss(observer_state, fs, loss_mode, rss_cfg, loss_cfg):
    def objective(omega):
        S = reconstruct_candidate_sources(omega, observer_state, rss_cfg)
        loss, _ = compute_rss_loss(S, fs, loss_mode, loss_cfg)
        return loss

    omega_star = run_existing_pso(objective, rss_cfg)
    S_hat = reconstruct_candidate_sources(omega_star, observer_state, rss_cfg)
    loss_star, diag = compute_rss_loss(S_hat, fs, loss_mode, loss_cfg)
    return omega_star, S_hat, loss_star, diag
```

Suggested loss names:

- `corr_zero_lag`;
- `harmonic_spectral_overlap`;
- `filtered_multilag_xcorr`;
- `chirp_tf_overlap`;
- `piecewise_sparse_innovation`.

The important architectural rule is that the quaternion/SO(3) reconstruction layer must not contain source-family conditionals. Only the loss layer should depend on the assumed source structure.

15 Validation Criteria for Deciding Whether a Specialized Loss is Useful

A specialized loss should not be considered successful merely because its own objective reaches a small value. Before using it in a paper, verify all of the following:

1. **Physical identification:** low loss correlates with low $d_{\mathrm{SO}(3)}(A, A_{\mathrm{true}})$ in synthetic loss-landscape tests.
2. **Source reconstruction:** it improves or at least preserves ground-truth correlation and angular RMSE relative to correlation-RSS on paired cases.
3. **Robustness:** improvement is not restricted to one or two favorable cases or one PSO seed.
4. **Structural validity:** the true source signals actually satisfy the property used by the loss. For example, true harmonic sources must have sufficiently distinct spectral support for a spectral-overlap loss to make sense.
5. **No hidden ground truth:** the loss must use only the candidate recovered signals and known experiment settings such as f_s and a justified analysis band. It must not use the true source signals, true frame, true frequencies, or generator parameters during optimization.
6. **Fair interpretation:** a source-family-specific loss is a structure-informed RSS variant because the family/property is assumed known. It should not be described as an automatically adaptive blind method unless a separate procedure is developed to choose the loss without oracle family labels.

16 Relationship Between the Four Losses

The proposed objectives can be interpreted as minimizing overlap in different representations:

harmonic:	overlap of global frequency-energy distributions,	(74)
filtered noise:	overlap/dependence across temporal lags,	(75)
chirp:	overlap of local time–frequency distributions,	(76)
piecewise transient:	overlap and nonsparsity of temporal innovations.	(77)

In every case, the geometry constrains the candidate signals to those obtainable through a physically valid $A \in \mathrm{SO}(3)$ transformation of the common quaternion vector part. The source prior determines which member of that admissible set is preferred.

17 Compact Algorithm Summary

Input: observer-angle matrix $\Theta_R \in \mathbb{R}^{3\times N}$, sample rate f_s, source-family label for specialized experiments, current RSS/PSO configuration, and loss configuration.

1. Convert every observer-angle triplet to $q_R(t) = (w_R(t), \boldsymbol{v}_R(t))$ using the existing ordered quaternion composition.
2. Initialize PSO exactly as in the existing RSS implementation.
3. For every candidate rotation vector $\boldsymbol{\omega}$:
   (a) compute $A(\boldsymbol{\omega})$ using Rodrigues' formula;
   (b) compute $\boldsymbol{v}_S(t) = A(\boldsymbol{\omega})\boldsymbol{v}_R(t)$ and $w_S(t) = w_R(t)$;
   (c) reconstruct the three candidate source-angle signals;
   (d) apply only the numerical preprocessing needed for loss evaluation;
   (e) compute the family-specific scalar loss;
   (f) return that scalar to PSO.
4. After the fixed PSO budget, return $\boldsymbol{\omega}^\star$, $A^\star$, and $\widehat{\mathbf{S}}$.
5. Evaluate the returned sources using exactly the same ground-truth metrics as the current validation notebook.
6. Store loss-specific diagnostics and, for synthetic data, SO(3) frame error.

18 Implementation Checklist for Codex

Before treating the new notebook as complete, verify:

Original correlation objective reproduces the existing RSS results after refactoring.
No quaternion, rotation-order, Rodrigues, source-angle, PSO, or data-generation logic has been changed merely to implement new losses.
Exact existing PSO boundary handling and fixed-seed behavior are preserved from the codebase.
All losses use only candidate recovered signals and permissible known settings; no ground-truth leakage enters optimization.
All divisions use small numerical epsilons and all inverse-trigonometric inputs are clipped where required.
Frequency/time–frequency losses handle angle wrapping consistently.
The same synthetic cases and frame transforms are reused for paired baseline/specialized comparisons.
Loss-landscape checks are run before full 200-case specialized experiments.

Every run saves its full loss configuration and seed metadata.
Existing notebook plots/metrics are reproduced, with additional loss-specific diagnostics added rather than replacing existing outputs.

19 Notes on Interpretation for the Journal Paper

These experiments are exploratory and need not all be included in the four-page IEEE Signal Processing Letters manuscript. Their immediate purpose is to determine whether the broader formulation

$$A^\star = \arg\min_{A\in \mathrm{SO}(3)} \mathcal{L}_{\mathcal{P}}(\mathbf{S}(A)) \tag{78}$$

is supported empirically by more than one identification criterion.

If the specialized losses work well, the short paper can state the general loss-agnostic formulation while retaining the correlation objective as the primary evaluated blind instantiation. One compact specialized-loss result could then be included only if it materially strengthens the argument and fits the page limit. The remaining experiments can be retained as internal validation, supplementary material if permitted, or a basis for a longer follow-up manuscript.

If a specialized loss does *not* improve recovery, that result is still informative. It would indicate that the corresponding intuitive signal property is not sufficiently identifying under the current synthetic generator, rather than invalidating the quaternion/SO(3) geometry itself.

A Family Definitions from the Current RSS Synthetic Study

For implementation alignment, the current manuscript describes the four waveform families before centering, normalization, amplitude scaling, and rotational composition as follows.

A.1 Harmonic

$$g_m^H(t) = \sin(2\pi f_{m,1} t + \phi_m) + 0.3\sin(2\pi f_{m,2} t - \phi_m), \tag{79}$$

with documented frequency ranges

$$f_{m,1} \in [0.2, 0.9] \text{ Hz}, \qquad f_{m,2} \in [1.0, 3.2] \text{ Hz}. \tag{80}$$

Because these ranges are shared by all three sources, distinct spectral support is not guaranteed in every case. This is why Section 11 is especially important for the harmonic loss.

A.2 Filtered noise

$$g_{m,n}^F = \alpha_m g_{m,n-1}^F + (1-\alpha_m)\xi_{m,n}, \qquad \xi_{m,n} \sim \mathcal{N}(0,1), \tag{81}$$

with

$$\alpha_m \in [0.88, 0.985]. \tag{82}$$

The substantial temporal memory in this range motivates evaluating cross-correlation over multiple lags rather than using only lag zero.

A.3 Chirp modulated

$$g_m^C(t) = [0.75 + 0.25\sin(2\pi f_{\mathrm{mod},m}\tau + \phi_m)] \tag{83}$$

$$\times \sin\left(2\pi\left[f_{0,m}\tau + \frac{f_{1,m} - f_{0,m}}{2T_{\mathrm{dur}}}\tau^2\right] + \phi_m\right), \tag{84}$$

where $\tau = t - t_0$ and the manuscript gives

$$f_{0,m} \in [0.1, 0.7]\ \mathrm{Hz}, \quad f_{1,m} \in [1.1, 4.2]\ \mathrm{Hz}, \quad f_{\mathrm{mod},m} \in [0.05, 0.30]\ \mathrm{Hz}. \tag{85}$$

This family motivates a time-resolved spectral criterion.

A.4 Piecewise transient

The current manuscript describes the family as a linear-interpolation component combined with a sum of Gaussian bursts,

$$g_m^P(t) = \sum_{r=1}^{R_m} b_{m,r} \exp\left[-\frac{1}{2}\left(\frac{t - c_{m,r}}{w_{m,r}}\right)^2\right] + \mathrm{lininterp}(t_j, v_{m,j})(t), \tag{86}$$

with 6–10 interpolation knots and 2–5 Gaussian bursts. This structure motivates sparse second-difference innovations and event-overlap analysis.

B Notation Summary

Symbol	Meaning
A	proper-orthogonal observer-to-source coordinate transformation
$\boldsymbol{\omega}$	three-parameter rotation vector used to represent A
$\boldsymbol{v}_R, \boldsymbol{v}_S$	quaternion vector-part coordinates in observer and source bases
w_R, w_S	frame-invariant scalar quaternion component
$\mathbf{S}(\boldsymbol{\omega})$	candidate recovered source-angle matrix
$\mathcal{L}_{\mathrm{corr}}$	original zero-lag correlation loss
$\mathcal{L}_{\mathrm{H}}$	harmonic spectral-overlap loss
$\mathcal{L}_{\mathrm{F}}$	filtered-noise multi-lag cross-correlation loss
$\mathcal{L}_{\mathrm{C}}$	chirp time–frequency overlap loss
$\mathcal{L}_{\mathrm{P}}$	piecewise sparse-innovation/event-overlap loss
$B_{ij}^{(H)}$	pairwise harmonic spectral Bhattacharyya overlap
$C_{ij}^{(F)}$	weighted multi-lag squared cross-correlation energy
$\overline{B}_{ij}^{(C)}$	pairwise mean time–frequency Bhattacharyya overlap
H_i	Hoyer sparsity of source-i second-difference innovations
$O_{ij}^{(P)}$	pairwise event-envelope overlap for piecewise signals
$d_{\mathrm{SO(3)}}$	geodesic rotation error between estimated and true frames

C Source Basis and Explicit Unknowns

This implementation reference was formulated from the supplied *Formulation_of_RSS.pdf* and the current *IEEE_Signal_Processing_Letters_RSS.pdf*, with the BSS comparison note used only as contextual validation material. The source documents specify the RSS geometry, current correlation objective, PSO parameter values, and synthetic family definitions. They do *not* specify the exact PSO boundary operator used by the software, and the current validation notebook itself was not supplied. Those details must therefore be copied from the existing codebase rather than inferred from this document.

The specialized losses in Sections 5–8 are new proposed extensions for experimental testing. They should be treated as hypotheses about useful source structure until the loss-landscape and reconstruction experiments demonstrate that they identify the true source frame on the current validation cases.